INDIA INNOVATES 2026

Team Name:

Orion : The Ambition Factory

ORION

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Project Name:

TRILOKA Systems : Autonomous Tri-Domain Robotic Exploration System for Civil-Defence Maritime Operation

Members Name and Affiliation:

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PROBLEM STATEMENT

- **The Challenge :** Monitoring 7500+ km of Indian coastline and 5,000+ large dams is currently labor-intensive and high-risk and suffers from “Data Blind Spots”.

- **Civilian Risk :** Manual Inspection of aging dams is dangerous, and slit-heavy waters makes visual monitoring impossible.

- **Defence Risk :** Harbours are vulnerable to “assymetric threats”like aubmerged mines or unauthorized divers that standard radar cannot detect.

- **The Gap :** Current solutions are single-domain (only air or only water), expensive and often rely on imported technology.

SOLUTION

- **The Approach :** A unified tri-domain platform– **Varuna** (Surface),**Garud** (Sky), and **Matsya** (Underwater)– operating as a single coordinated ecosystem.

- **Civilian Use :** Autonomous bathymetry(depth mapping) and structural health monitoring of reservoirs.

- **Defence Use :** Stealthy littoral reconnaissance and “Mine Counter-Measure” (MCM) operations.

- **Innovation :** A Passive-Flooding Moon-Pool for silent underwater deployement and a Gyroscopic Balancing System for high-stability operations in rough coastal waters.

ARCHITECTURE

- **System Design :** A hierarchical "Mothership-Daughter" configuration where the **NVIDIA Jetson Nano Super** acts as the central AI brain.

- **Interaction :** Coordinated deployment: The moon-pool flaps open via servo-control only when Varuna reaches the GPS-locked inspection coordinates.

- **Data Flow :**
  - **Garud (Aerial) :** Sends real time reconnaissance data via ELRS to Varuna for path planning.
  - **Matsya (Sub) :** Streams high bandwidth inspection video through a fiber optic tether to the central hub.
  - **Varuna (Surface) :** Fuses all sensor data (LIDAR, IR camera, GPS) and transmits telemetry to the operator via LoRa or 5G mesh networks.

TECHNOLOGY USED

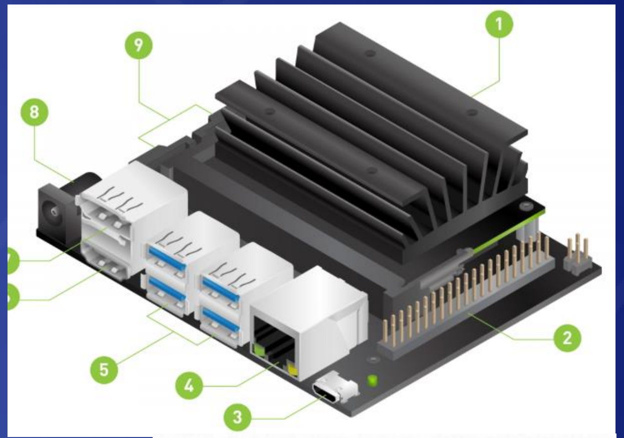
- **Computing :** NVIDIA Jetson Nano Super for “Edge AI” threat/crack detection.

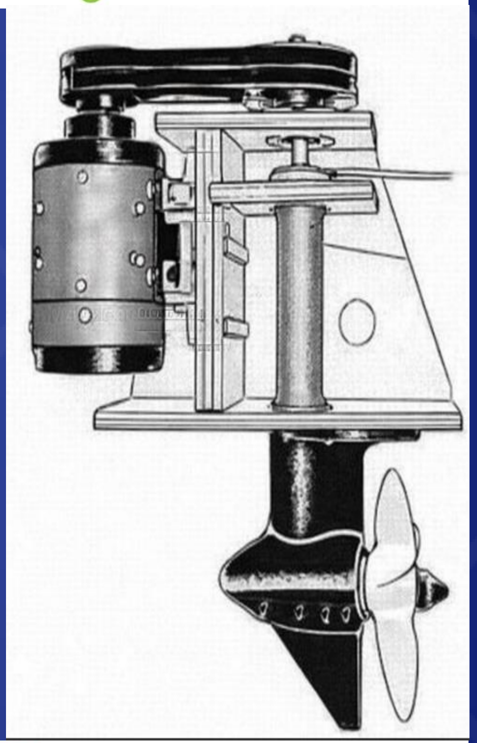
- **Propulsion :** 3000W BLDC motors with a custom 90-degree bevel gear transmission for high-efficiency marine thrust/rapid tactical response.

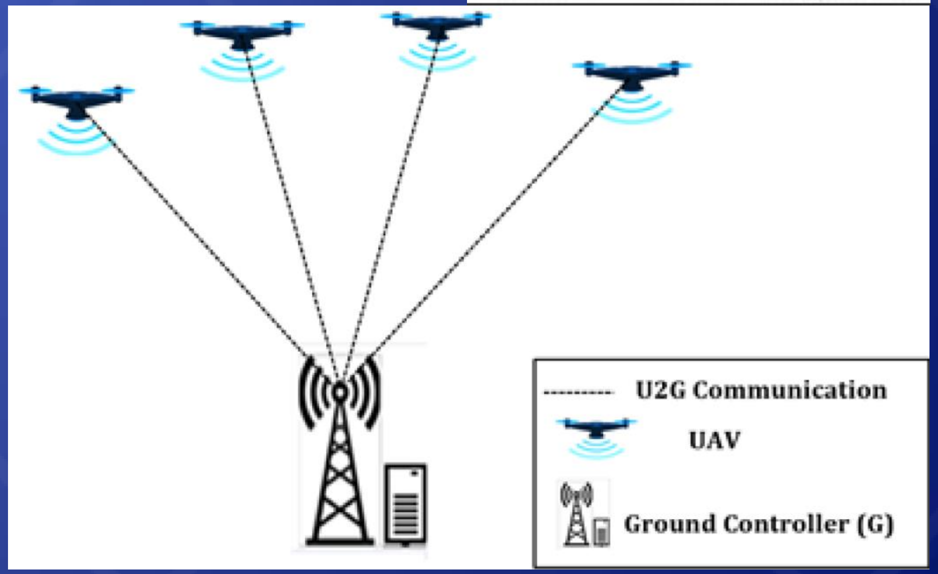
- **Sensors :** Waveshare camera for low-visibility infrared underwater scanning and spectral imaging for aerial environmental monitoring and fuel trails from stealth craft.

- **Communication :** ELRS for drone control, Fiber-optic tether for the ROV, and LoRa/5G for long-range vessel telemetry.

- **Power :** 22,000 mAh LiPo battery supported by integrated Flexible Solar Panels for extended mission endurance.







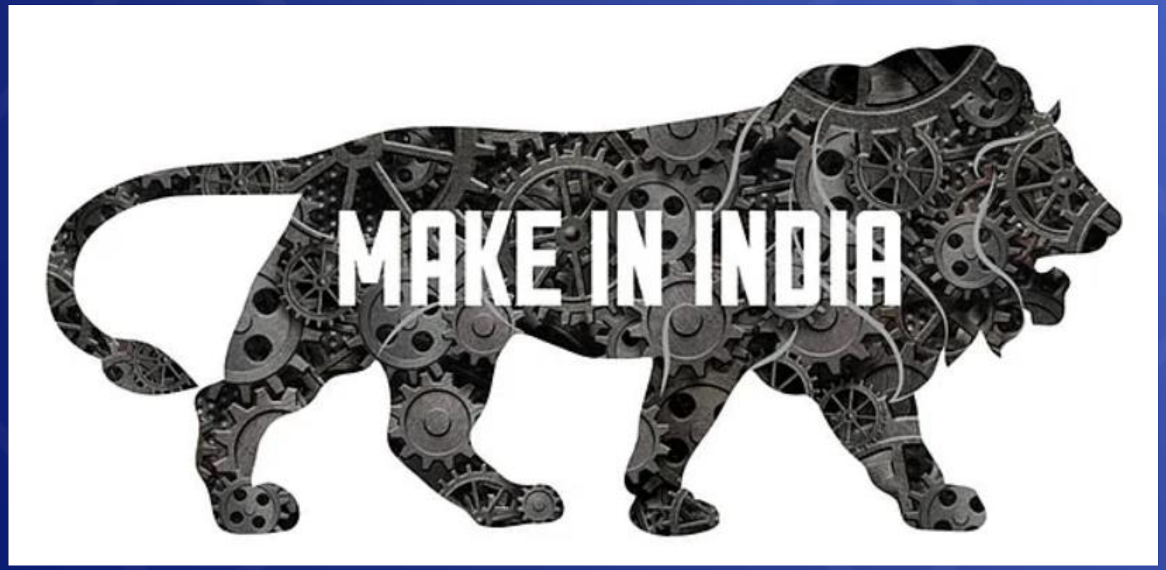
FEATURE / USP

- **Tri-Domain Integration :** The only low-cost platform providing simultaneous data from the sky, water surface and underwater in one mission.

- **Passive Moon-Pool System :** A unique low-complexity deployment mechanism that uses water pressure to equalize levels, ensuring reliable sub-surface launches.

- **Strategically Attritable :** With a hull cost under ₹12,000, these units are "expendables", allowing deployment in high-risks zone(toxic spills or minefields) where manned assets cannot go.

- **Indigenous Manufacturing :** Designed for local production using fiberglass-reinforced EPS, making it a "Make In India" solution for marine robotics.



REFERENCES / LINKS

- **Technical Research :** Based on Hindu cosmology naming conventions (Varuna-Garud-Matsya) and autonomous catamaran stability studies using gyroscopic precision.
- **Strategic Alignment :** Designed to meet the goals of both the National Dam Safety Act (civil) and the iDEX/Swavlamban naval initiatives (Defence).

THANK YOU

TRILOKA Systems: Redefining aquatic exploration and maritime security for a safer and more sustainable India